

CRF State Summary v11

(φ, e) System Session — April 2026

Maturity: 8.0/8 Theory · 7.5/8 Empirical gate open

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Abstract

Three new results since State Summary v10.

Algebraic impossibility (new): $f(8) = \sqrt{\varphi}$ is proved to be *not* algebraically exact by Lindemann–Weierstrass. The 0.004% gap is irreducible and transcendental in origin.

K21 is ε_0 -free (proved): The ε_0 cancels exactly in K21's expansion, making it a pure function $f(n)$ of n alone. Upgraded from Hypothesis to Proved.

D_0 from φ : K21 implies $D_0 \approx \sqrt{\varphi} \cdot G(8)$ where $G(8) = 7\sqrt{19} \cdot 62/2560$ — a direct relation between the resolution scale and the golden ratio (gap 0.004%).

The $(\varphi, e) \rightarrow (n, \varepsilon_0) \rightarrow$ everything chain stands, with all gaps now understood as transcendental (not algebraic) in origin.

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1 Primitive Chain

$$\delta \rightarrow \mathcal{P} \rightarrow \mathcal{C} \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow \mathcal{R} \rightarrow t$$

Master chain: $(\varphi, e) \rightarrow (n = 8, \varepsilon_0) \rightarrow D_0, K^*, \beta, \kappa, \sigma_\Psi, \psi_{\text{grad}}, F, H_a, H_b, T_{\text{avg}}, \dots$

2 New This Session

Result	Status	Significance
$f(8) = \sqrt{\varphi}$ impossible	Proved	Lindemann–Weierstrass; gap is irreducible
K21 is ε_0 -free	Proved	Exact algebraic cancellation
$D_0 = \sqrt{\varphi} \cdot G(8)$	Hypothesis	0.004%; direct D_0 - φ relation
$(\varphi, e) \rightarrow (n, \varepsilon_0)$ complete	Structural	0.005% overall; no free parameters

3 The (φ, e) System

Two constants determine CRF

Layer 1 (ε_0 -free, geometry): Define $f(n) = 4n^2 D_0(n)(n+2)/[(n-1)\sqrt{2n+3}(n^2-2)]$. Then $f(n) = \sqrt{\varphi}$ selects $n = 8$ as the unique integer solution (gap 0.004%). The gap is transcendently irreducible (Lindemann–Weierstrass).

Layer 2 (dynamical, given $n = 8$): K20 quadratic $C(1 + \varphi_{\text{var}})e \varepsilon_0^2 + (1 + \varphi_{\text{var}})e \varepsilon_0 - 2D_0^2/(n(n-1)) = 0$ gives $\varepsilon_0 = 0.007550$ (gap 0.005%).

Consequence: $n = 8$, $d_s = 3$ (from $n = 2^{d_s}$), and all simulator constants from (ε_0, n) algebraically. Zero free parameters.

4 Complete Status Table

Result	Status	Note
<i>The (φ, e) pinning</i>		
$f(n)$ is ε_0 -free (K21)	Proved	Exact cancellation, new
$f(8) = \sqrt{\varphi}$ impossible	Proved	Lindemann–Weierstrass, new
$f(8) \approx \sqrt{\varphi}$ pins $n = 8$	Hypothesis	0.004%, transcendental gap
K20 derives ε_0	Hypothesis	0.005%
$D_0 = \sqrt{\varphi} \cdot G(8)$	Hypothesis	0.004%, new
$n = 2^{d_s}$ (binary Born rule)	Physical arg	exact
<i>Resolution lattice</i>		
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$K^* = D_0/n$ (K1)	Derived	0.43%
$\beta D_0 = \ln n$ (K11)	Derived	0.005%
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Hypothesis	0.047%
K17, K18, K19 ratios	Derived	exact
<i>Wave fields</i>		
$p = (n + 1)/(n + 2)$	Derived	exact
$\sigma_\Psi = \varepsilon_0(n + 2)/\sqrt{2n + 3}$	Derived	0%
$\varphi_{\text{var}} = 1/2 - 1/n^2$	Derived	2.3%
$\psi_{\text{grad}} = \sigma_\Psi / \ln \varphi$ (K8)	Hypothesis	0.016%
$F = 2D_0^2/(n(n - 1)\varepsilon_0 e)$ (K20)	Hypothesis	0.005%
$\text{Cov}(\psi_{\text{grad}}, \Delta H) = 0$	Derived	exact via K3
<i>β-chain</i>		
$(H_a + H_b)/2 = \sqrt{d_s}$ (K3)	Derived	0.011%
$T_{\text{avg}} = 2D_0/((n - 1)\varepsilon_0)$	Derived	exact (K1)
β formula (measured inputs)	4 Derived	0.44%

Open

5 Key Numbers v11

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8 = 2^3$	$N = 500$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5378$
$T_{\text{avg}} = 35.57$	$\varphi = 1.61803$	$e = 2.71828$	$H_a = 1.7925$	$H_b = 1.6717$
$\varphi_{\text{var}} = 0.4844$	$\sigma_{\Psi} = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$H_u = 2.055$	$p = 9/10$
$G(8) = 7\sqrt{19} \cdot 62/2560$	$D_0/\sqrt{\varphi} = 0.73900$	$f(8) = 1.27208$	$\sqrt{\varphi} = 1.27202$	

6 Open Gaps

Priority 1: $m(m-1) = 1$ energy balance

The last free Hypothesis. Path: find $f(\delta)$ in $m(m-1) = \beta\kappa/f(\delta)$ such that $m = \varphi$ without s_{max} dependence.

Priority 2: $n = 2^{d_s}$ formal proof

Derive from δ -chain that the Born-rule mode count is exactly $n = 2^{d_s}$.

Priority 3: Why $f(8) \approx \sqrt{\varphi}$ to 0.004%?

Proved NOT exact by Lindemann–Weierstrass. But why so close? The answer involves how $n = 2^{d_s}$ forces $D_0(8) \approx \sqrt{\varphi} \cdot G(8)$ — a consequence of the log-dual structure $n = 2^{d_s}$ and the golden ratio's role in 3D geometry.

7 Documents This Session

Document	Content
CRF_K20K24.pdf	K20–K24; e enters CRF; $n = 8$ pinned
CRF_PhiESystem.pdf	Impossibility proof; D_0 from φ ; two-layer clean statement
CRF_StateSummary_v11.pdf	This document

$f(8) = \sqrt{\varphi}$ to 0.004%.
 Not exact. Not accidental.
 The gap is transcendental: $e^{-1/8}$ and $\sqrt{\varphi}$ live in different mathematical worlds,
 yet at $n = 8$, they come within 0.004% of each other.
 φ selects the symmetry class. e calibrates the noise.
 Together they build the world from δ .